Compressor off-design report: summary and diagnostics

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Rotor-Stator  |  run: Max_eta_Compressor_design_rs_no_last_stator

Rotor-Stator  |  PR_des = 1.4939, eta_des = 0.9312  |  correlations: polynomial, Howell factor 2.00

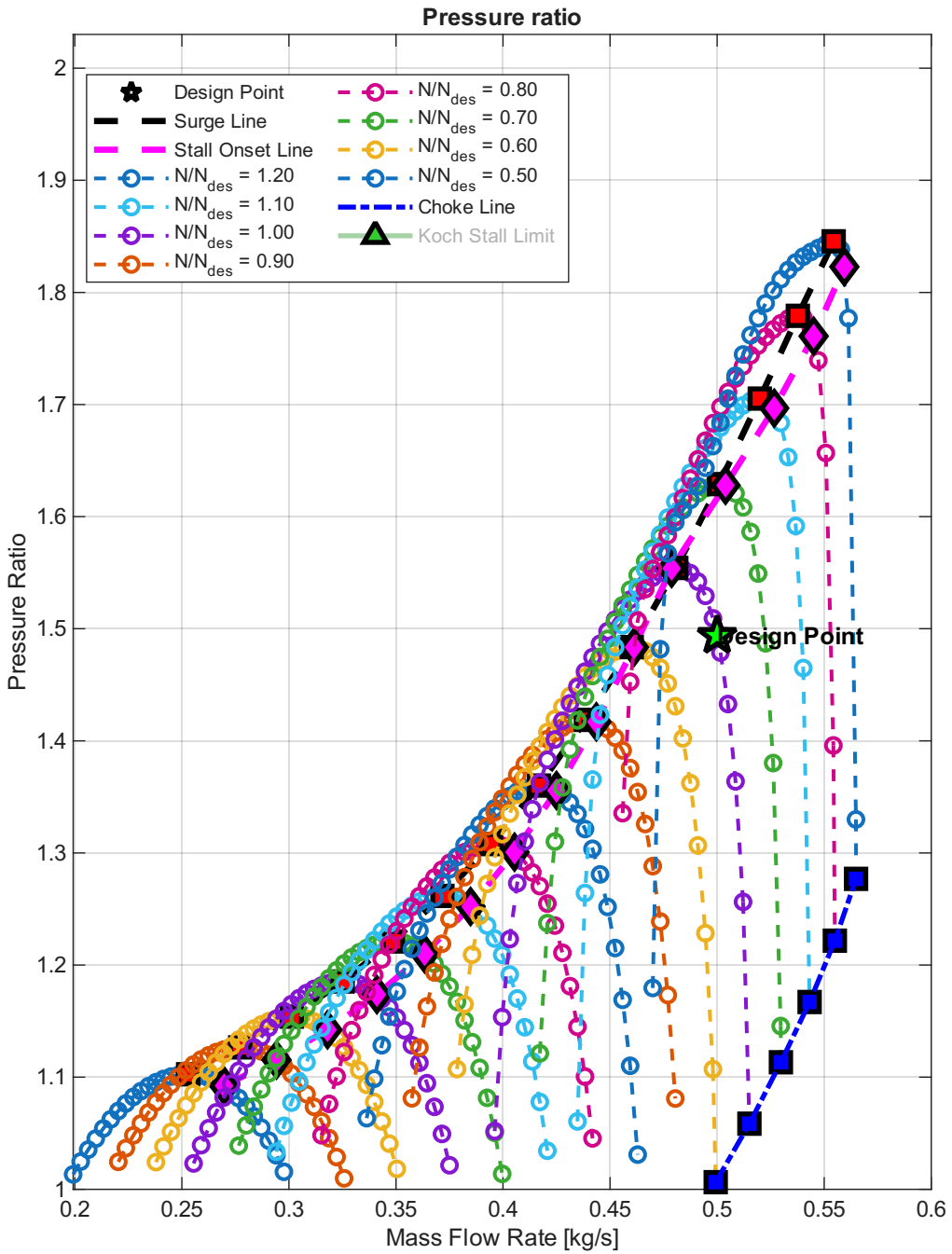
STABILITY (design speed): first limit = peak at m = 0.4809 kg/s (PR 1.554); stall margin SM = 8.2 %
   peak of the speed line (zero slope): m = 0.4809 kg/s | Howell stall onset: m = 0.4787 kg/s
   Koch stage limit: m = 0.5018 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.5151 kg/s (Rotor 2 exit (relative)) -> choke margin 3.0 %, operating range (m_choke - m_lim)/m_choke = 6.6 %
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.377 -> surge if the installation has V_p/(A_c L_c) > 4.51 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Stator 1 (omega/omega_des = 1.87), lowest de Haller Rotor 1 (0.79). Details: Tables > Blade
      rows.

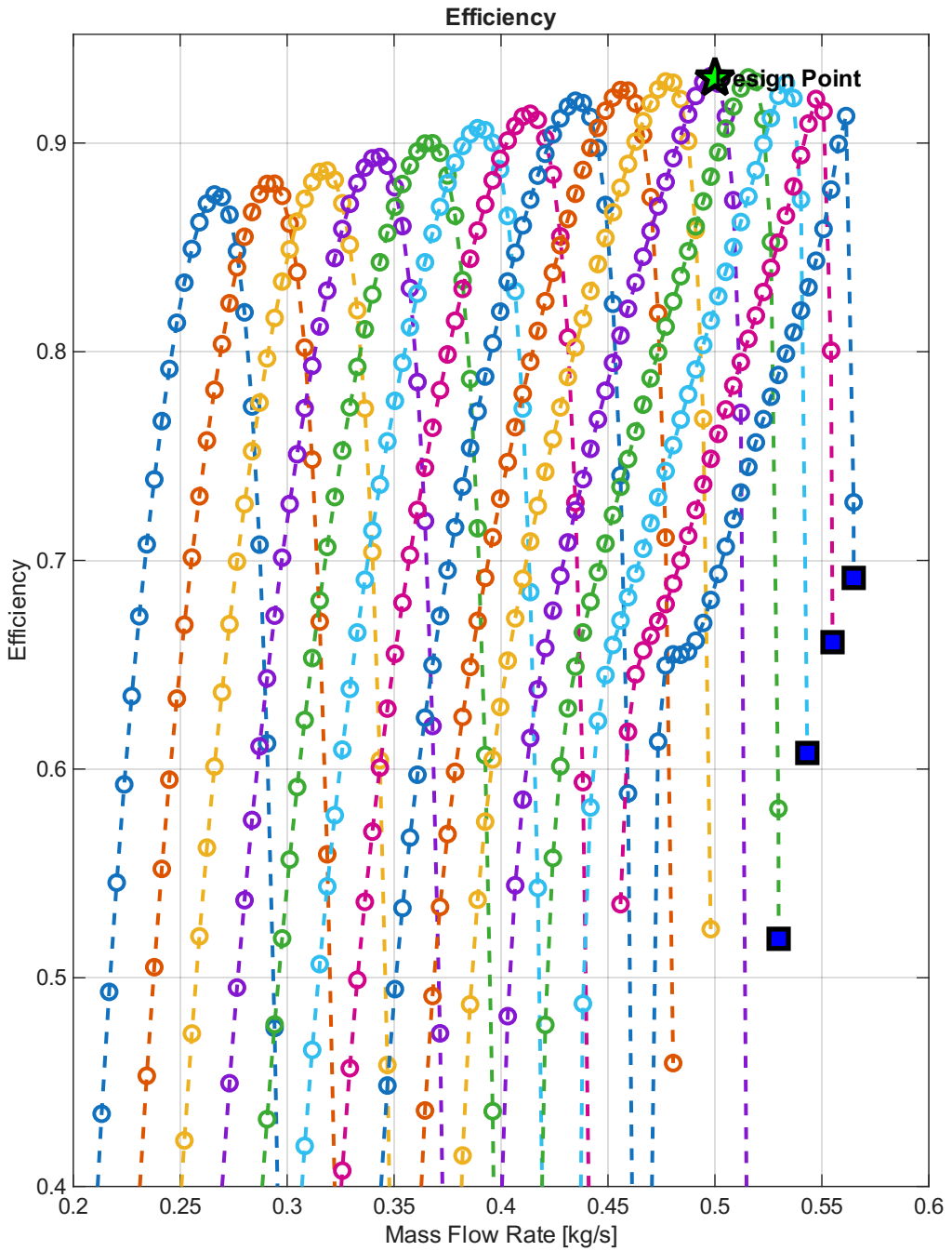
KOCH STAGE-LOADING INDICATOR (never a stability limit): INDICATIVE
   Indicative: near Koch's data. Read the ratio as a trend (more or less loaded than a typical stage), not as a limit. Howell and the peak set the
      limit.
   Koch line: drawn on 9 of 15 speed lines (N/N_des 0.80 to 1.20).
   why: Stage 1 (R1+S1): L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: Stage 2 (R2 only): L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Stage 1 (R1+S1): BEYOND THE LIMIT AT DESIGN (1.04)
     - RAISE the solidity x1.42 (stage L/g2 0.53 -> 0.76): R1 0.50 -> 0.71, S1 0.50 -> 0.71; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 18% at the same velocities (lower loading psi or degree of reaction, or a larger flow
       coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.95
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.97
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2) -> indicative only
  Stage 2 (R2 only): BEYOND THE LIMIT AT DESIGN (1.03)
     - RAISE the solidity x1.39 (stage L/g2 0.53 -> 0.74): R2 0.50 -> 0.70; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 17% at the same velocities (lower loading psi or degree of reaction, or a larger flow
       coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.94
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.96
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2) -> indicative only
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Inputs of this run (.mat data and settings)

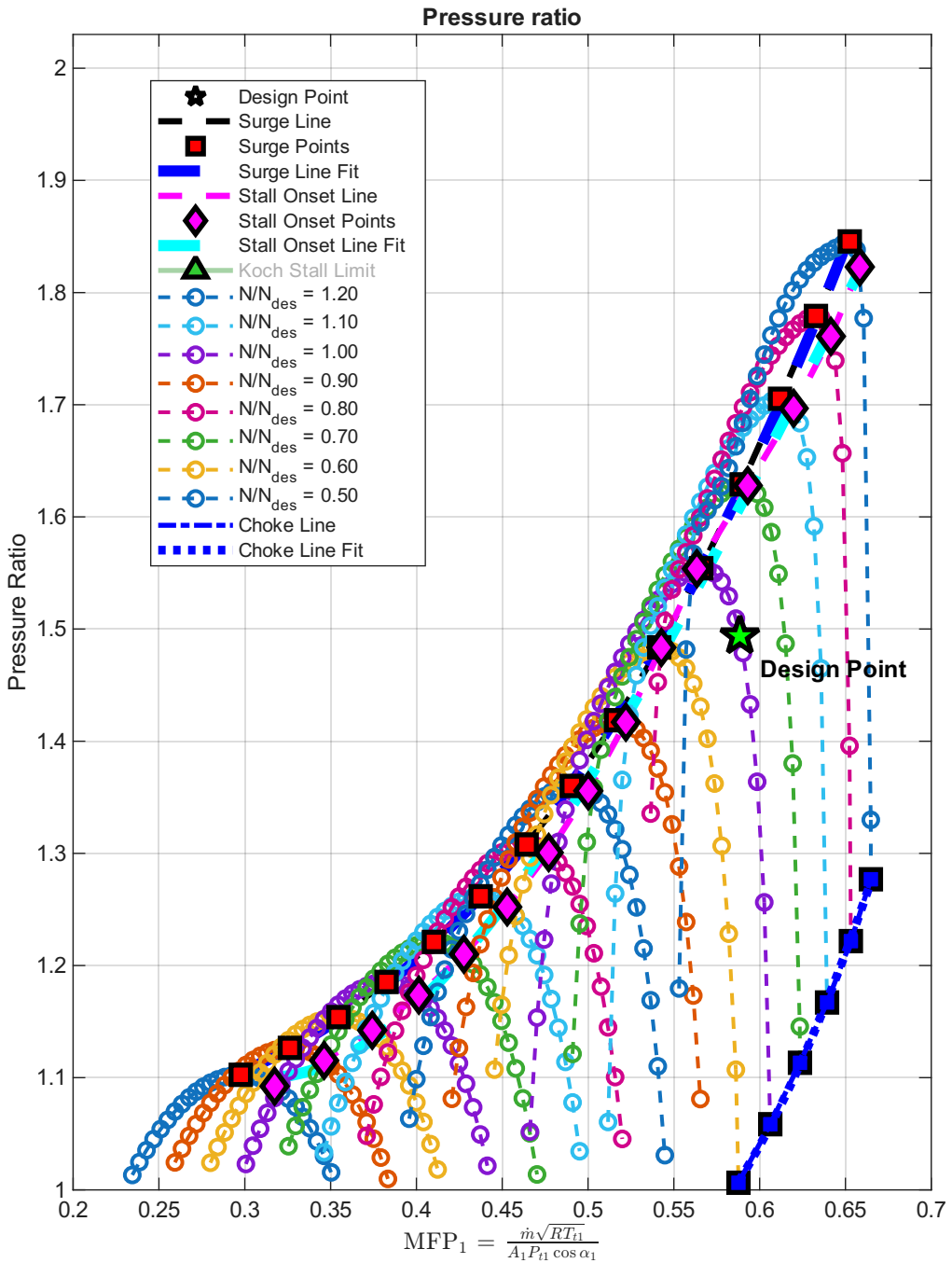
Variable	Value	Unit	Source
Configuration	Rotor-Stator		
Input source	.mat: Max_eta_Compressor_design_rs_...		
alpha1	30.3676	deg	.mat: Max_eta_Compressor_design_rs_...
n_stages	2	-	.mat: Max_eta_Compressor_design_rs_...
A_vec	0.00283128 0.00262358 0.00243314 0....	m^2	.mat: Max_eta_Compressor_design_rs_...
beta1_dash	50.3693	deg	.mat: Max_eta_Compressor_design_rs_...
beta2_dash	18.077	deg	.mat: Max_eta_Compressor_design_rs_...
m_dot	0.5	kg/s	.mat: Max_eta_Compressor_design_rs_...
w_loss_rotor	0.0324827	-	.mat: Max_eta_Compressor_design_rs_...
deviation	12.2907	deg	.mat: Max_eta_Compressor_design_rs_...
incidence	-6.04865	deg	.mat: Max_eta_Compressor_design_rs_...
deflection	13.953	deg	.mat: Max_eta_Compressor_design_rs_...
w_loss_stator	0.0266822	-	.mat: Max_eta_Compressor_design_rs_...
U	274.775	m/s	.mat: Max_eta_Compressor_design_rs_...
sigma_rotor	0.500466	-	.mat: Max_eta_Compressor_design_rs_...
sigma_stator	0.500466	-	.mat: Max_eta_Compressor_design_rs_...
chord	0.01	m	.mat: Max_eta_Compressor_design_rs_...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	129	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

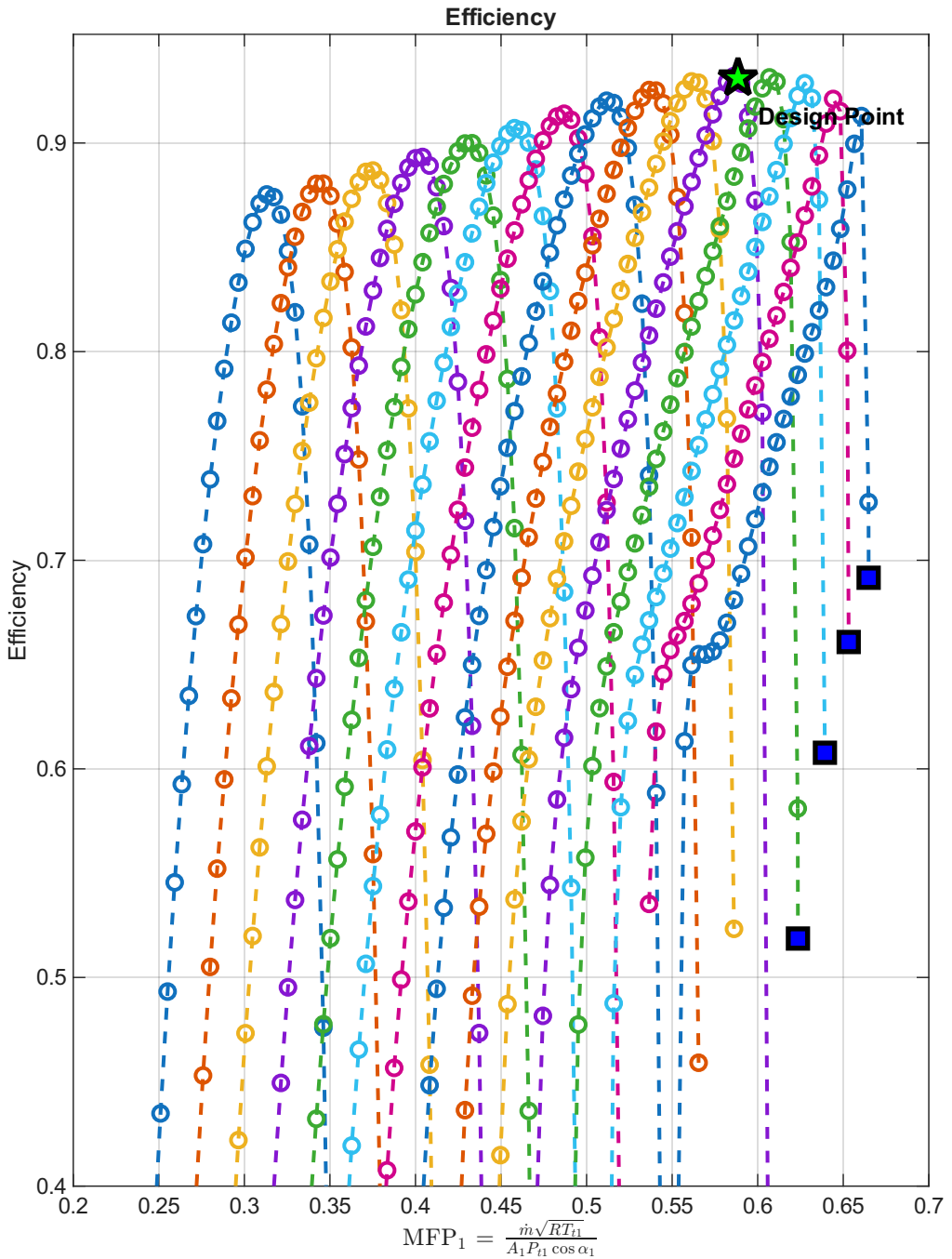
Standard (dimensional) map - Rotor-Stator



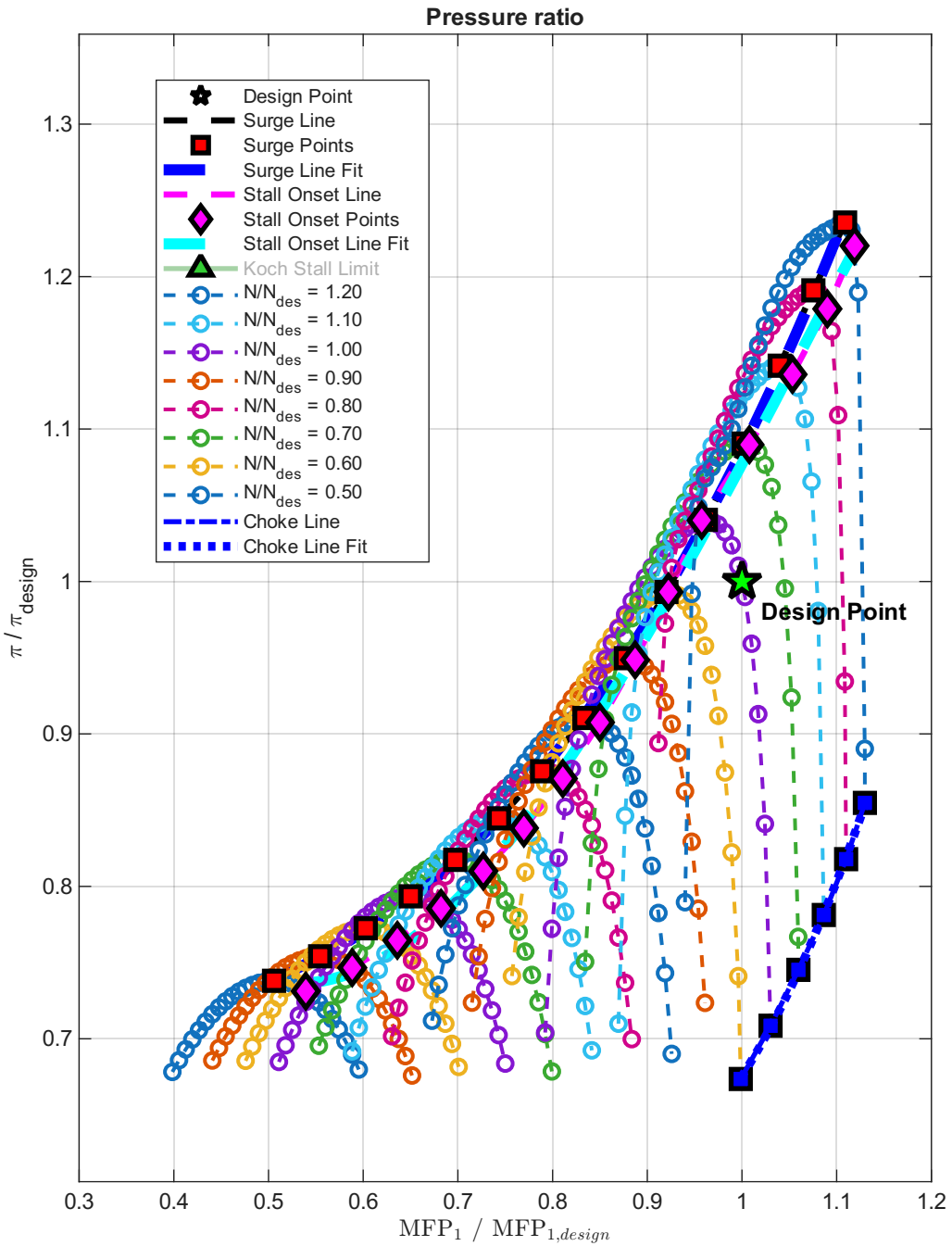


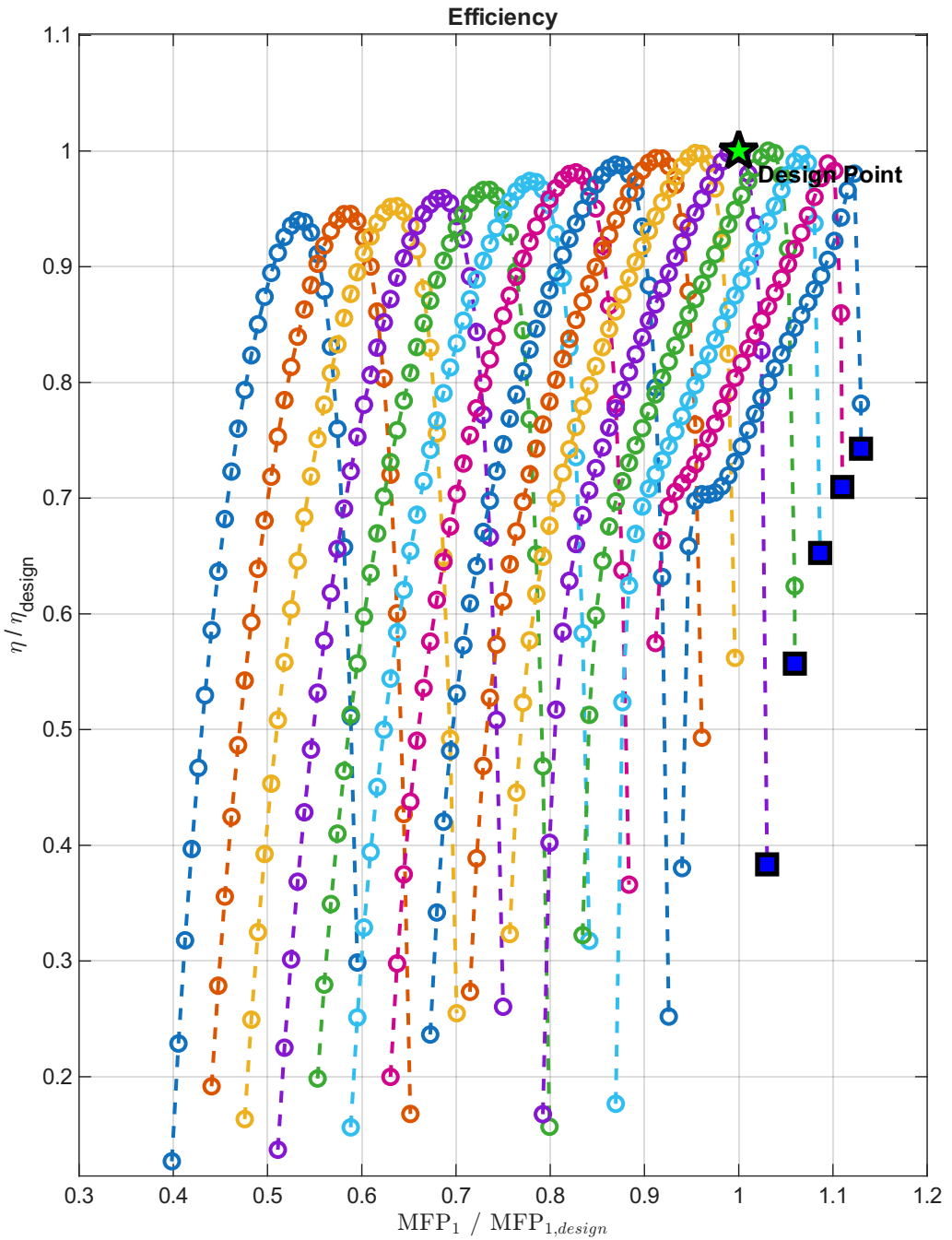
Non-dimensional map - Rotor-Stator



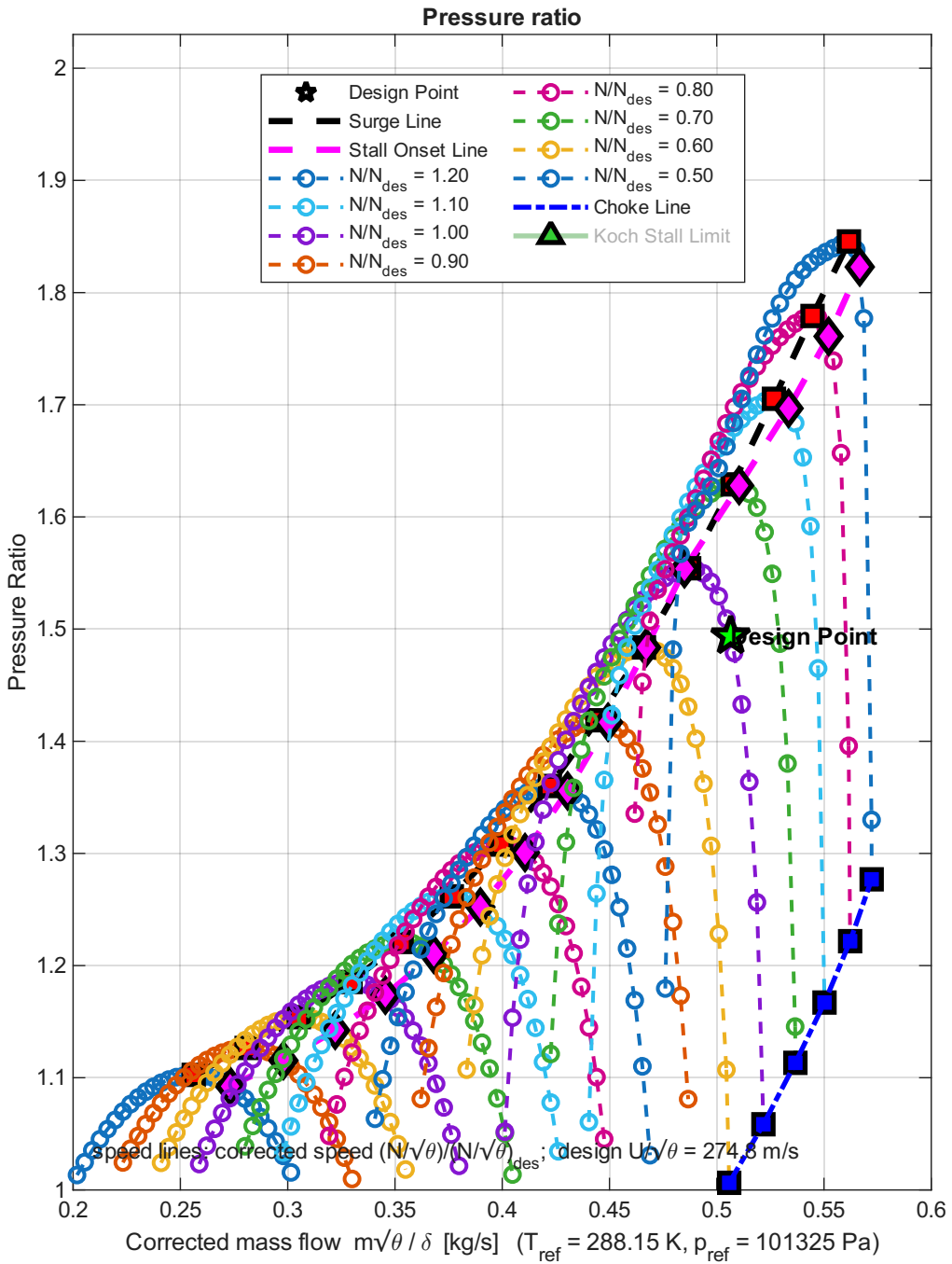


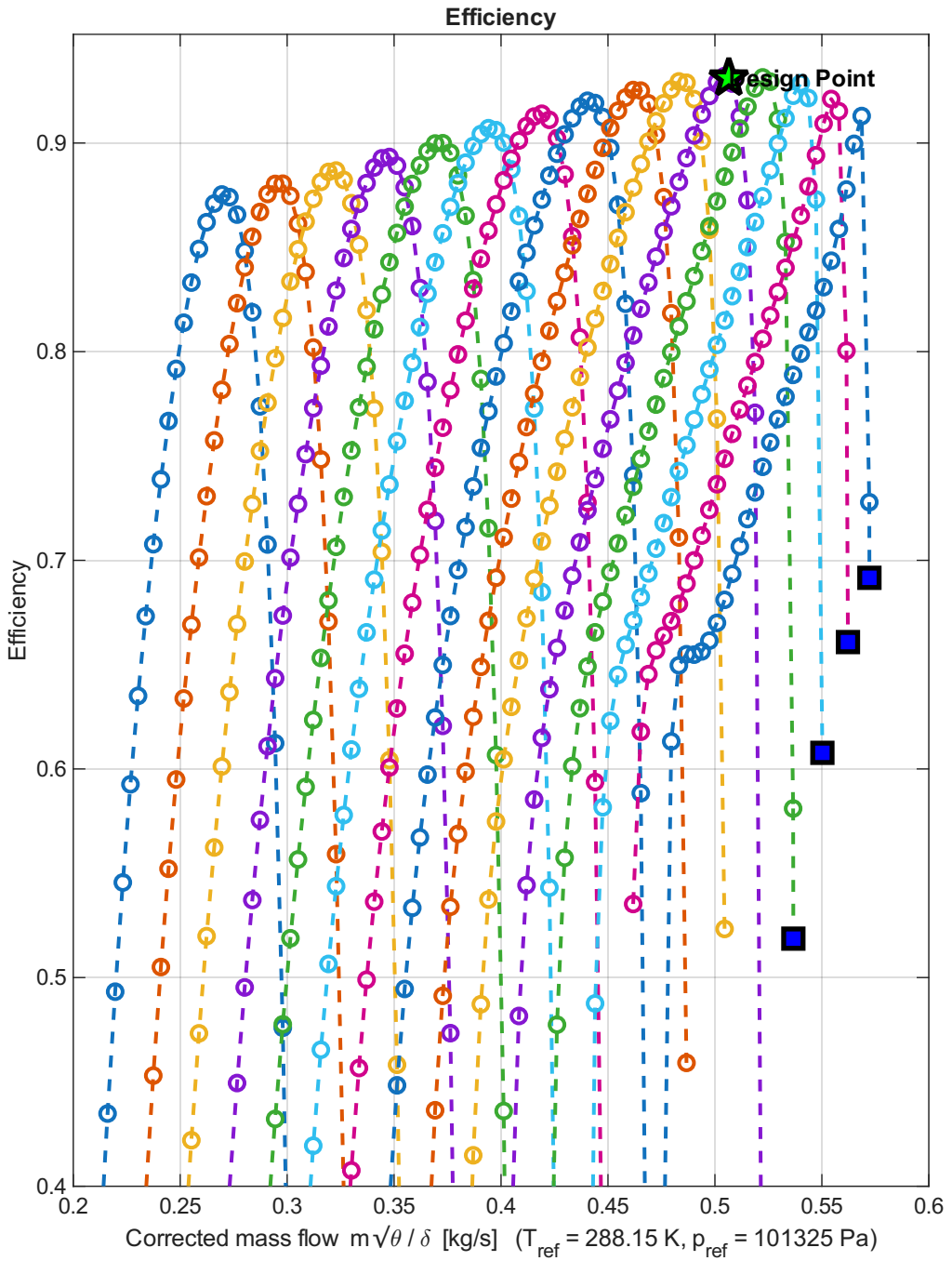
Relative non-dimensional map - Rotor-Stator





Corrected map - Rotor-Stator





Fitted boundary equations (surge, stall onset, choke)

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==== Non-dimensional Compressor Map: Pressure Ratio - Rotor-Stator ====
Surge line:
Surge_PR = 3.2981*MFP^3 + -0.0142*MFP^2 + -0.2191*MFP + 1.0844   [R^2 = 0.99986]
Stall-onset line (Howell):
StallOnset_PR = -9.5265*MFP^3 + 18.2144*MFP^2 + -8.5848*MFP + 2.2948   [R^2 = 0.99891]
Choke line:
Choke_PR = 13.4404*MFP^2 + -13.3675*MFP + 4.2214   [R^2 = 0.99964]
==== Relative Non-dimensional Compressor Map: Pressure Ratio - Rotor-Stator ====
Surge line:
(PR/PR_design) = 0.4499*x^3 + -0.0033*x^2 + -0.0863*x + 0.7259   (x = MFP/MFP_design)   [R^2 = 0.99986]
Stall-onset line (Howell):
(PR/PR_design) = -1.2995*x^3 + 4.2222*x^2 + -3.3817*x + 1.5361   (x = MFP/MFP_design)   [R^2 = 0.99891]
Choke line:
(PR/PR_design) = 3.1155*x^2 + -5.2657*x + 2.8258   (x = MFP/MFP_design)   [R^2 = 0.99964]
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Design point

Quantity	Value
Configuration	Rotor-Stator
Pressure ratio PR_des	1.4939
Isentropic efficiency eta_des	0.93116
Mass flow m_des [kg/s]	0.5
MFP_1,des	0.58846
Corrected mass flow [kg/s]	0.50649
Inlet Tt1 [K] / Pt1 [Pa]	288 / 100000
Correlation set	polynomial
Howell factor	2

Choke

N/N_des	m_choke [kg/s]	MFP_1	PR	Eff	Choking row	Inside compressor region (PR>1)
0.5	0.33942	0.3995	0.626		Rotor 2 exit (relative)	no
0.55	0.35744	0.4207	0.66		Rotor 2 exit (relative)	no
0.6	0.37581	0.4423	0.696		Rotor 2 exit (relative)	no
0.65	0.39413	0.4639	0.734		Rotor 2 exit (relative)	no
0.7	0.41225	0.4852	0.773		Rotor 2 exit (relative)	no
0.75	0.43025	0.5064	0.815		Rotor 2 exit (relative)	no
0.8	0.44809	0.5274	0.859		Rotor 2 exit (relative)	no
0.85	0.46571	0.5481	0.906		Rotor 2 exit (relative)	no
0.9	0.48283	0.5682	0.956		Rotor 2 exit (relative)	no
0.95	0.49934	0.5877	1.006	0.055	Rotor 2 exit (relative)	yes
1	0.51508	0.6062	1.059	0.357	Rotor 2 exit (relative)	yes
1.05	0.52979	0.6235	1.113	0.519	Rotor 2 exit (relative)	yes
1.1	0.54319	0.6393	1.167	0.608	Rotor 2 exit (relative)	yes
1.15	0.55499	0.6532	1.222	0.661	Rotor 2 exit (relative)	yes
1.2	0.56489	0.6648	1.277	0.692	Rotor 2 exit (relative)	yes

Stall onset (Howell)

N/N_des	m_stall [kg/s]	PR	Eff	Stalling row
0.5	0.26968	1.092	0.874	Rotor 1
0.55	0.2942	1.115	0.88	Rotor 1
0.6	0.31802	1.142	0.887	Rotor 1
0.65	0.34111	1.174	0.893	Rotor 1
0.7	0.36339	1.21	0.899	Rotor 1
0.75	0.38483	1.252	0.903	Rotor 1
0.8	0.40538	1.301	0.906	Rotor 1
0.85	0.42497	1.356	0.906	Rotor 1
0.9	0.44358	1.417	0.903	Rotor 1
0.95	0.46116	1.484	0.896	Rotor 1
1	0.47873	1.554	0.887	Stator 1
1.05	0.50392	1.628	0.903	Rotor 2
1.1	0.52667	1.697	0.914	Rotor 2
1.15	0.54508	1.761	0.915	Rotor 2
1.2	0.55941	1.823	0.909	Rotor 2

Peak (zero slope)

N/N_des	m_peak [kg/s]	PR	Eff	m_stall - m_peak [kg/s]	First limit on throttling
0.5	0.25292	1.103	0.838	0.01676	stall onset (Howell)
0.55	0.27725	1.127	0.843	0.01694	stall onset (Howell)
0.6	0.30132	1.154	0.85	0.0167	stall onset (Howell)
0.65	0.3251	1.185	0.856	0.016	stall onset (Howell)
0.7	0.34857	1.221	0.863	0.01482	stall onset (Howell)
0.75	0.3717	1.262	0.87	0.01314	stall onset (Howell)
0.8	0.39445	1.308	0.877	0.01092	stall onset (Howell)
0.85	0.41681	1.36	0.883	0.00817	stall onset (Howell)
0.9	0.43871	1.419	0.888	0.00487	stall onset (Howell)
0.95	0.46011	1.484	0.892	0.00105	stall onset (Howell)
1	0.48091	1.554	0.894	-0.00218	peak (zero slope)
1.05	0.50093	1.629	0.894	0.00299	stall onset (Howell)
1.1	0.51992	1.705	0.89	0.00675	stall onset (Howell)
1.15	0.53769	1.779	0.883	0.00739	stall onset (Howell)
1.2	0.55444	1.846	0.879	0.00497	stall onset (Howell)

Koch limit

N/N_des	m_Koch [kg/s]	PR	Eff	Limiting stage
0.5	NaN	NaN	NaN	not reached
0.55	NaN	NaN	NaN	not reached
0.6	NaN	NaN	NaN	not reached
0.65	NaN	NaN	NaN	not reached
0.7	NaN	NaN	NaN	not reached
0.75	NaN	NaN	NaN	not reached
0.8	0.32312	1.106	0.364	Stage 2 (R2 only)
0.85	0.42926	1.349	0.915	Stage 1 (R1+S1)
0.9	0.45742	1.385	0.926	Stage 1 (R1+S1)
0.95	0.48085	1.428	0.928	Stage 1 (R1+S1)
1	0.50176	1.477	0.928	Stage 1 (R1+S1)
1.05	0.52126	1.516	0.921	Stage 2 (R2 only)
1.1	0.53844	1.542	0.907	Stage 2 (R2 only)
1.15	0.55259	1.564	0.888	Stage 2 (R2 only)
1.2	0.56384	1.584	0.865	Stage 2 (R2 only)

Stability limit & Greitzer B

N/N_des	First limit	m_lim [kg/s]	PR	U [m/s]	a_p [m/s]	U/(2a_p)	(V_p/(A_c L_c))_crit	B	Beyond the limit
0.5	stall onset	0.26968	1.092	137.4	345.1	0.199	16.15	NaN	-
0.55	stall onset	0.2942	1.115	151.1	346.2	0.2182	13.44	NaN	-
0.6	stall onset	0.31802	1.142	164.9	347.5	0.2372	11.37	NaN	-
0.65	stall onset	0.34111	1.174	178.6	349	0.2559	9.77	NaN	-
0.7	stall onset	0.36339	1.21	192.3	350.6	0.2743	8.51	NaN	-
0.75	stall onset	0.38483	1.252	206.1	352.5	0.2924	7.49	NaN	-
0.8	stall onset	0.40538	1.301	219.8	354.5	0.31	6.66	NaN	-
0.85	stall onset	0.42497	1.356	233.6	356.8	0.3273	5.98	NaN	-
0.9	stall onset	0.44358	1.417	247.3	359.4	0.3441	5.41	NaN	-
0.95	stall onset	0.46116	1.484	261	362.1	0.3604	4.93	NaN	-
1	peak	0.48091	1.554	274.8	364.8	0.3766	4.51	NaN	-
1.05	stall onset	0.50392	1.628	288.5	367.2	0.3928	4.15	NaN	-
1.1	stall onset	0.52667	1.697	302.3	369.3	0.4092	3.82	NaN	-
1.15	stall onset	0.54508	1.761	316	371.4	0.4254	3.54	NaN	-
1.2	stall onset	0.55941	1.823	329.7	373.6	0.4413	3.29	NaN	-

Blade rows

Operating point (N = 1)	m [kg/s]	Blade row	Incidence i [deg]	(i - i_des)/deflection	Loss omega	omega/omega_des	M_in (row frame)	M_out (row frame)	de Haller V_out/V_in	Deviation [deg]	Watch
Design point	0.5	Rotor 1	-6.05	0	0.0325	1	0.75	0.611	0.829	12.29	
Design point	0.5	Stator 1	-6.05	0	0.0267	1	0.737	0.601	0.829	12.29	
Design point	0.5	Rotor 2	-6.05	0	0.0325	1	0.725	0.592	0.829	12.29	
First limit (peak)	0.48091	Rotor 1	-3.3	0.197	0.0564	1.736	0.738	0.57	0.788	12.88	near Howell
First limit (peak)	0.48091	Stator 1	-3.34	0.194	0.0499	1.87	0.717	0.557	0.791	12.87	near Howell
First limit (peak)	0.48091	Rotor 2	-3.13	0.209	0.0594	1.829	0.703	0.545	0.79	12.95	near Howell
Near choke (0.99 m_choke)	0.50993	Rotor 1	-7.57	-0.109	0.0398	1.226	0.758	0.646	0.865	12.23	
Near choke (0.99 m_choke)	0.50993	Stator 1	-8.35	-0.165	0.0433	1.625	0.751	0.657	0.886	12.24	
Near choke (0.99 m_choke)	0.50993	Rotor 2	-9.73	-0.264	0.0746	2.298	0.748	0.692	0.932	12.28	